

PROJECT REPORT

Predictive Analytics for Early Clinical Deterioration

Using SAS Visual Analytics for Patient Risk Monitoring

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Course / Tool Focus	Data Science project using SAS Visual Analytics
Project Theme	Early warning analytics for clinical deterioration within a 12-hour horizon

1. Summary

This project analyses a simulated hospital cohort to identify patients likely to experience early clinical deterioration. The central objective is to detect deterioration risk within the next 12 hours so that hospitals can shift from reactive intervention to proactive monitoring. The analysis combines patient-level demographics and baseline risk with hourly vital-sign and laboratory time-series data.

The project uses three raw data streams: `patients.csv`, `vitals_timeseries.csv`, and `labs_timeseries.csv`. These streams were merged into a unified hourly dataset containing patient profiles, physiological observations, laboratory markers, risk labels, and deterioration outcomes. SAS Visual Analytics was used to build interactive dashboards, compare risk segments, identify warning signals, and translate findings into operational monitoring recommendations.

Key findings show that nearly 1 in 5 patients deteriorated, while near-term 12-hour risk was around 3.1%. Risk was not evenly distributed: older patients, patients with very high comorbidity, high baseline risk, abnormal respiratory rate, low oxygen saturation, elevated heart rate, infection markers, kidney stress indicators, and transfer status showed stronger deterioration patterns. Only 12.5% of patients were classified as High Priority, making targeted enhanced monitoring realistic for hospital operations.

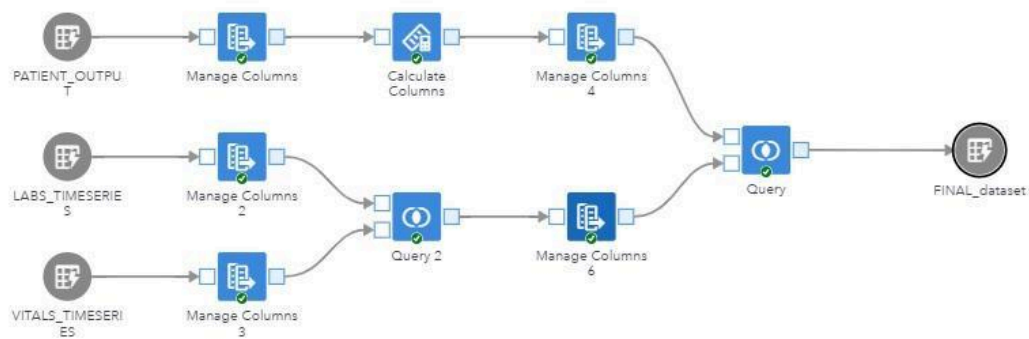


Figure 1. Data pipeline via SAS workflow.

2. Introduction

Clinical deterioration refers to a worsening of a patient's condition that may require urgent clinical intervention, escalation of care, transfer, or intensive monitoring. In hospital settings, deterioration is often preceded by changes in vital signs, laboratory values, oxygen support, nurse alerts, and patient mobility. Detecting these warning signs early is important because delay in recognition can reduce the time available for treatment.

The project focuses on predictive analytics and visual storytelling rather than only descriptive reporting. The purpose is to transform raw hospital data into useful decision support: identify who is at higher risk, understand which factors contribute to deterioration, and recommend how hospitals can prioritize monitoring resources.

The work is structured around four main components: project overview and scope, dataset selection and justification, methodological rigor with exploratory data analysis, and visual analytics for decision-makers.

3. Problem Statement

Hospitals need a practical method to identify patients likely to deteriorate within the next 12 hours so that clinicians can intervene earlier and allocate monitoring resources efficiently. The project addresses the following analytical questions:

- How large is the deterioration problem in the patient population?
- Which patient groups deteriorate most based on age, comorbidity, admission type, and baseline risk?
- Do high-risk or high-priority patients deteriorate earlier than others?
- Which vital signs and laboratory markers show the strongest association with deterioration?
- Is deterioration sudden or gradual across the 72-hour monitoring window?

- Can risk segmentation support targeted monitoring and staff allocation?

Business/clinical importance: The project is operationally relevant because nearly 1 in 5 patients deteriorated. This indicates that deterioration is not a rare event and that structured early monitoring can contribute to patient safety and hospital resource efficiency.

4. Data

4.1 Dataset Selection and Justification

The dataset was selected because it simulates a realistic hospital monitoring environment. It contains patient-level attributes, hourly vital signs, hourly laboratory markers, and deterioration outcomes across a 72-hour observation window.

Dataset	Content	Purpose in the project
patients.csv	10,000 unique patients; one row per admission; demographics, admission type, baseline risk, and comorbidity.	Used to define patient profile, baseline risk, and clinical priority groups.
vitals_timeseries.csv	Hourly vital signs: heart rate, respiratory rate, SpO2, blood pressure, oxygen support, mobility score, nurse alerts.	Used to identify physiological warning signs and changes over time.
labs_timeseries.csv	Hourly laboratory markers: WBC, lactate, creatinine, CRP, hemoglobin, sepsis risk score.	Used to assess infection, metabolic stress, kidney function, and systemic instability.
Final unified dataset	Merged hourly clinical records with patient information, vitals, labs, and target labels.	Used for SAS Visual Analytics dashboards, EDA, and risk storytelling.

4.2 Data Cleansing

The raw data streams were merged into a final dataset suitable for visual analytics and machine-learning style target labels. The following cleansing activities were required:

- Checked patient identifiers so that records from patients, vitals, and labs could be merged correctly.
- Aligned hourly observations using patient_id and time-related fields to avoid mismatched patient timelines.
- Removed or filtered unusable records that could distort hourly monitoring analysis.
- Handled missing or invalid clinical readings so that trend charts and box plots did not produce misleading results.

- Standardised categorical labels such as baseline risk group, risk segment, age group, comorbidity group, admission type, and deterioration status.
- Verified that each final record represented a meaningful patient-hour observation for dashboard analysis.

4.3 Data Transformation

The project required transformation from separated raw tables into a unified analytical table. Key transformation steps were:

- 1. Import** patients.csv, vitals_timeseries.csv, and labs_timeseries.csv into the analytics workflow.
- 2. Prepare** each dataset separately by checking data types, identifiers, categorical fields, and missing values.
- 3. Join** patient-level records with hourly vitals using patient_id.
- 4. Join** laboratory time-series values using patient_id and the corresponding hourly time field.
- 5. Create** or preserve target labels such as deterioration_event and deterioration_within_12h.
- 6. Create** grouped variables for analysis: age group, comorbidity group, baseline risk group, and monitoring priority segment.
- 7. Aggregate** patient-hour data where needed to compare patient groups, deterioration rates, and median deterioration times.
- 8. Build** calculated measures such as deterioration rate, mean risk scores, total patients, and time-to-deterioration summaries.

4.4 Data Enrichment

Data enrichment was achieved by combining static patient information with dynamic time-series signals. Important enrichment outcomes included:

- Monitoring priority segments: Low, Medium, and High Priority groups for targeted monitoring.
- Risk comparison measures: deterioration rate by baseline risk, comorbidity, age group, and priority segment.
- Time-based deterioration measures: distribution of deterioration time and median time by priority segment.
- Combined clinical features: respiratory rate, SpO2, heart rate, lactate, creatinine, CRP, hemoglobin, sepsis risk score, oxygen support, nurse alerts, and mobility score.

5. Analysis

5.1 Population-Level Deterioration and Monitoring Priority

The dashboard summary showed 10,000 total patients, 418,000 hourly observations in the final analysis, 19% of patients deteriorating, and 3.1% near-term risk. This indicates that the problem is large enough to justify systematic monitoring but targeted enough to manage through priority segmentation.

Only 12.5% of patients were classified as High Priority. This is an important operational insight because it means enhanced monitoring does not need to be applied equally to all patients. Hospitals can focus more intensive monitoring on a smaller group while still tracking the broader population.



Figure 2. Key performance indicators: 10,000 total patients, 418,000 hourly observations, 19% patients deteriorating, and 3.1% near-term risk.

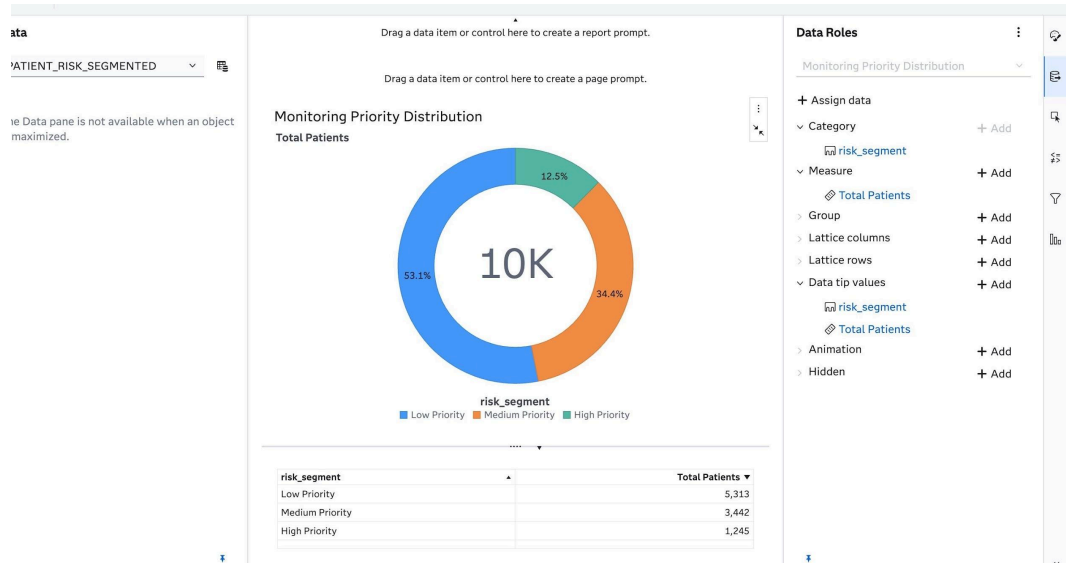


Figure 3. Monitoring Priority Distribution — 53.1% Low Priority (5,313 patients), 34.4% Medium Priority (3,442), and 12.5% High Priority (1,245). Only a small subset requires enhanced monitoring.

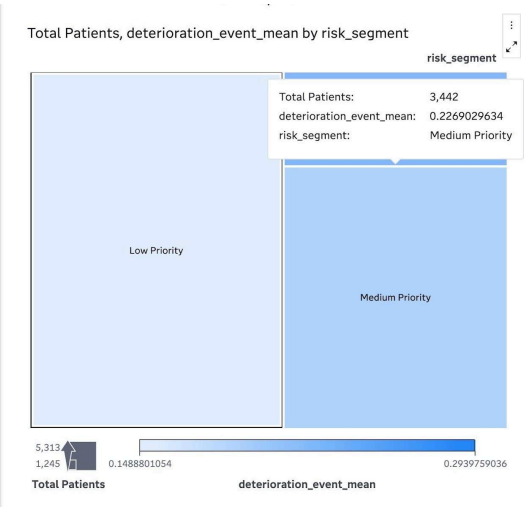


Figure 4. Treemap of Total Patients and deterioration_event_mean by risk_segment — tile size represents patient volume; colour intensity reflects deterioration rate. High Priority has the highest rate (0.29) despite the smallest population.

5.2 Patient Profile Risk: Age, Comorbidity, and Baseline Risk

The analysis found that deterioration risk is not evenly distributed across patient groups. The highest-risk segment was patients aged 80+ with very high comorbidity, reaching approximately 37% deterioration risk. This suggests that age and comorbidity together provide meaningful context for clinical vulnerability.

Baseline risk score also separated low-risk and high-risk patients effectively. Low Risk patients had approximately 12.6% deterioration, while High Risk patients reached approximately 26.6%.

Risk Dimension	Finding	Interpretation
Age + Comorbidity	80+ patients with very high comorbidity had the highest risk (~37%).	Older and more clinically complex patients require stronger monitoring attention.
Baseline Risk Group	High Risk patients deteriorated more than twice as often as Low Risk patients.	Admission risk scores are useful for prioritizing patients early.
Monitoring Segment	High Priority patients had the highest deterioration rate (~29.4%).	Priority segmentation supports focused resource allocation.
Timing	High Priority patients deteriorated ~4.5 hours earlier than Low Priority patients.	Early review is especially important during the first 24 hours.

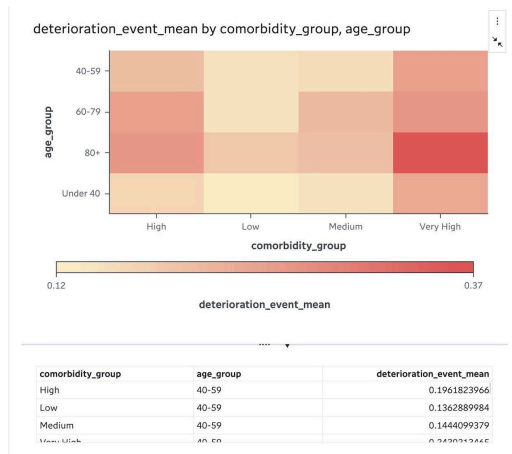


Figure 5. Deterioration event mean heatmap by comorbidity group and age group — the darkest cell (80+, Very High comorbidity) reaches 0.37, confirming that age and comorbidity together drive the highest deterioration rates.

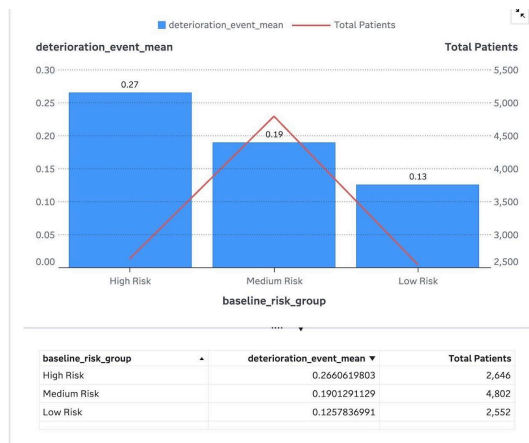


Figure 6. Deterioration event mean and total patients by baseline risk group — High Risk patients have the highest deterioration rate (0.27), while the Low Risk group is the smallest population (2,552 patients). The red line shows patient volume declining as risk increases.

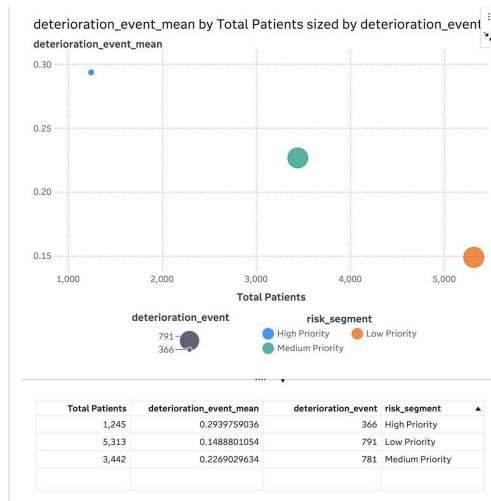


Figure 7. Bubble chart of deterioration_event_mean by total patients, sized by deterioration count and coloured by risk_segment — High Priority (blue) sits at a high deterioration rate (~0.29) with a smaller population (1,245); Low Priority (orange) has a large population (5,313) but a lower rate (~0.15).

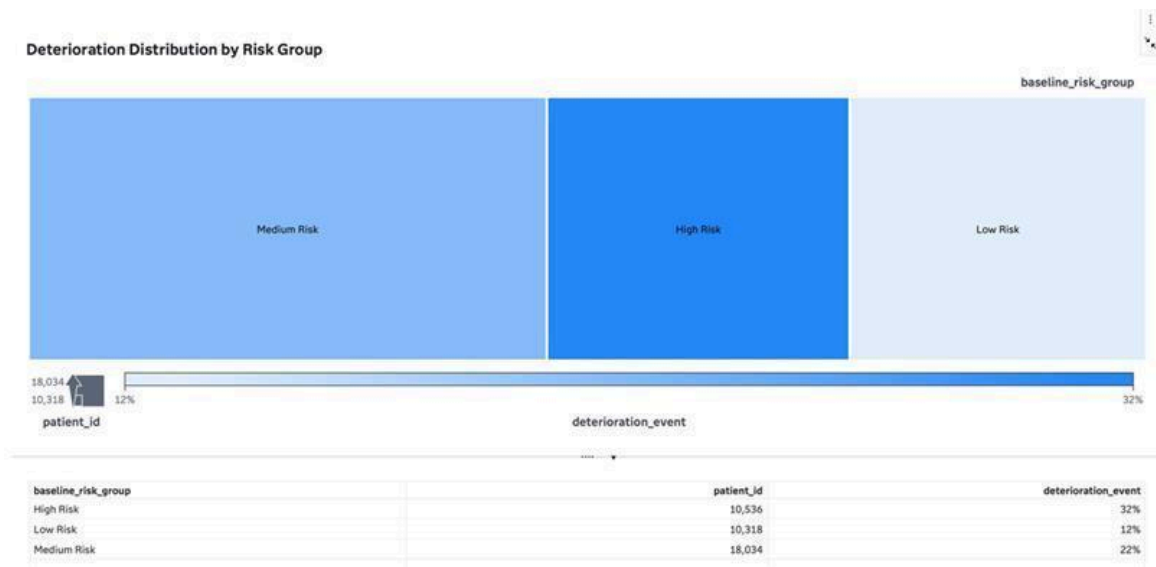


Figure 26. Deterioration Distribution by Risk Group (treemap) — tile area represents patient count; colour intensity reflects deterioration rate. High Risk patients (10,536) have the highest deterioration rate (32%), followed by Medium Risk (18,034 patients, 22%) and Low Risk (10,318 patients, 12%).

5.3 Timing of Deterioration

Timing was a key part of the analysis. High Priority patients deteriorated around 4.5 hours earlier by median time compared with Low Priority patients — 21.5 hours versus 26 hours. This means that early monitoring should not wait until visible deterioration becomes obvious. The first 24 hours after admission are particularly important, especially for patients already classified as High Priority.

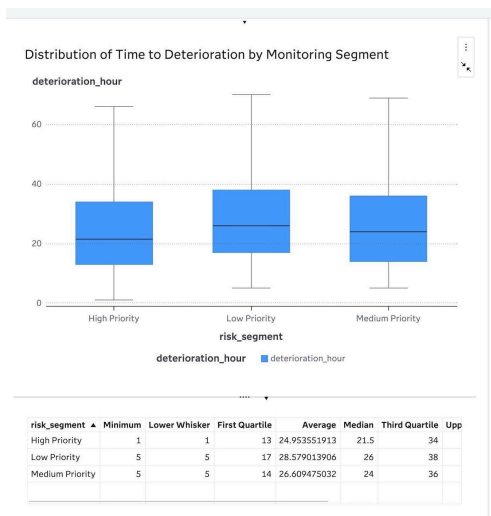


Figure 8. Distribution of Time to Deterioration by Monitoring Segment — High Priority patients have the lowest median deterioration hour (21.5h), followed by Medium Priority (24h) and Low Priority (26h), confirming that higher-risk patients deteriorate earlier.



Figure 9. Mean deterioration hour and 12-hour deterioration probability by baseline risk group — High Risk patients show the highest near-term risk (0.22) and the shortest mean deterioration hour (~26h), while Low Risk patients have the lowest near-term risk (~0.09) and later deterioration (~28h).

5.4 Vital Signs Associated with Deterioration

The vital-sign analysis showed that respiratory rate and oxygen saturation were the strongest warning signals. Deteriorating patients had much higher respiratory rates and lower SpO2 values than stable patients. Heart rate also increased among deteriorating patients, but heart rate alone was not sufficient.

Vital Sign	Observed Pattern	Clinical Meaning
Respiratory rate	Deteriorating patients breathed much faster; RR rose from ~20 to 38 breaths/min across 72 hours.	Strongest single vital-sign predictor in the visual analysis.
SpO2	Deteriorating patients showed lower oxygen saturation; SpO2 fell from ~94% to 74% across 72 hours.	Second strongest signal; indicates worsening respiratory/oxygenation status.
Heart rate	Average heart rate increased from ~84 BPM (stable) to ~108 BPM (deteriorating).	Useful signal, but not sufficient alone as many deteriorating patients remain in the normal zone.
Combined HR + RR	Patients with both high heart rate and high respiratory rate were more likely to deteriorate.	Combined indicators improve risk interpretation compared with one signal alone.

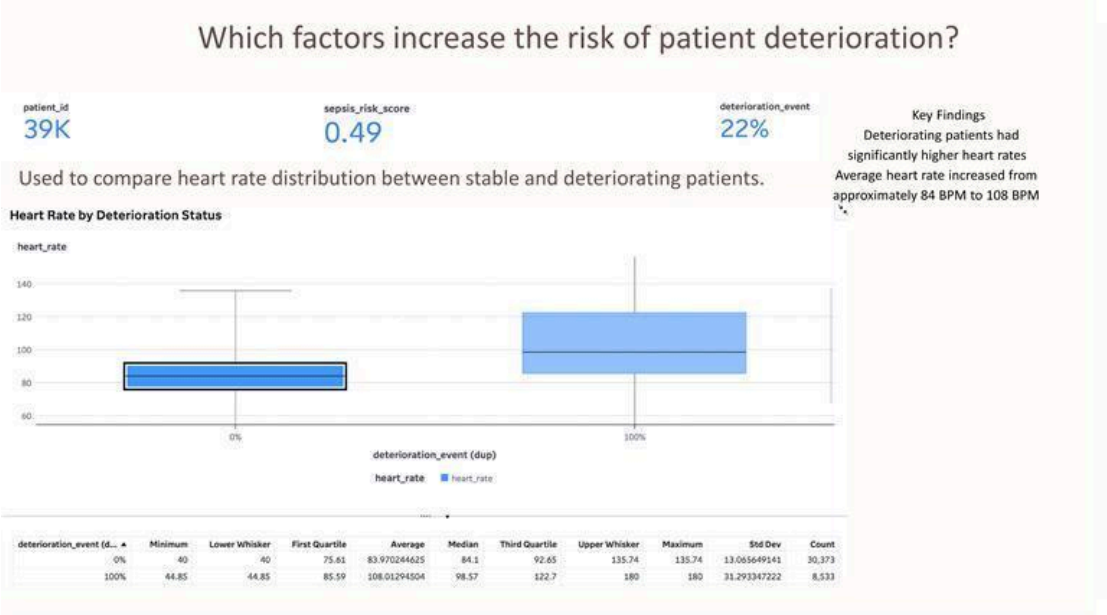


Figure 10. Heart Rate by Deterioration Status dashboard — KPIs show 39K observations, average sepsis risk score 0.49, and 22% deterioration rate. Deteriorating patients (100%) have a substantially higher median heart rate (~98.6 bpm) versus stable patients (0%, ~84 bpm), with a wider IQR and higher maximum.

Respiratory Rate by Deterioration Status

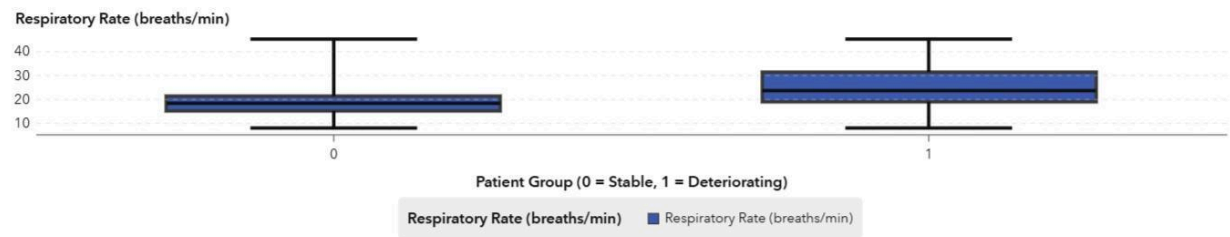


Figure 11. Respiratory Rate by Deterioration Status — deteriorating patients (group 1) show a higher median RR and wider range.

SpO2 by Deterioration Status

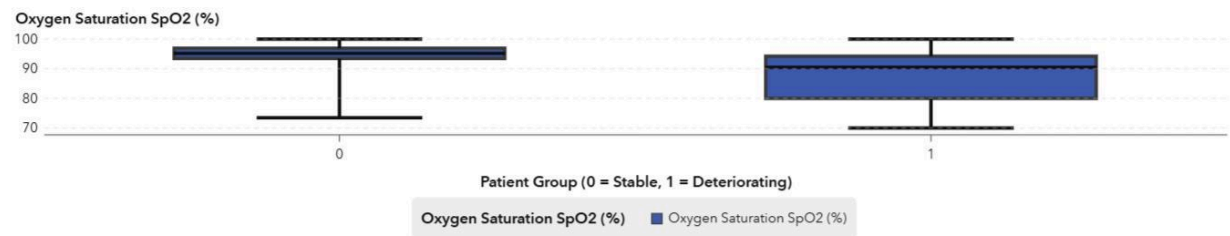


Figure 12. SpO2 by Deterioration Status — stable patients (group 0) maintain higher oxygen saturation; deteriorating patients show a marked downward shift.

Combined HR & RR Risk – Stable vs Deteriorating Patients



Figure 13. Combined HR & RR Risk — stable (blue) vs deteriorating (green) patients. Deteriorating patients cluster at higher heart rate and respiratory rate combinations.

5.5 Gradual Deterioration Pattern — Time-Series Trajectories

The time-series charts show that deterioration is gradual rather than sudden. In deteriorating patients, respiratory rate rose steadily over the 72-hour window while oxygen saturation declined steadily. Stable patients remained comparatively flat. This pattern supports the use of trend-based monitoring instead of relying only on isolated single readings.

The early-warning analysis shows that the first 6 hours may contain only a small difference between stable and deteriorating patients — around 1.5 breaths/min. By hour 24 the gap increased to about 15 breaths/min, meaning valuable intervention time may already be lost by the time deterioration becomes visually obvious.

Average Heart Rate Trajectory by Deterioration Status

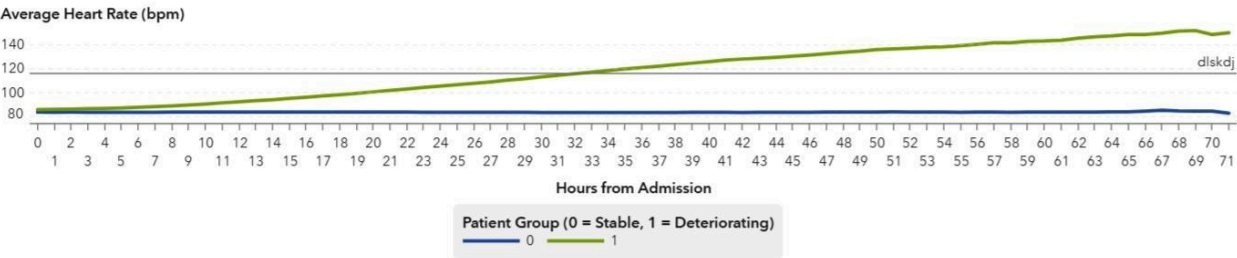


Figure 14. Average Heart Rate Trajectory by Deterioration Status — stable patients remain flat (~80 bpm); deteriorating patients rise steadily toward ~145 bpm by hour 71.

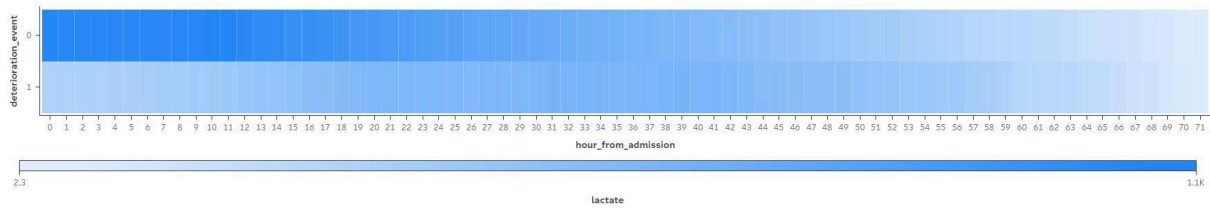


Figure 15. Systolic and Diastolic Blood Pressure Trajectory (0–71h) — systolic BP (blue) remains around 110 mmHg; diastolic BP (orange) tracks around 70 mmHg, both showing gradual decline in the later admission hours.

Respiratory Rate Trajectory by Deterioration Status

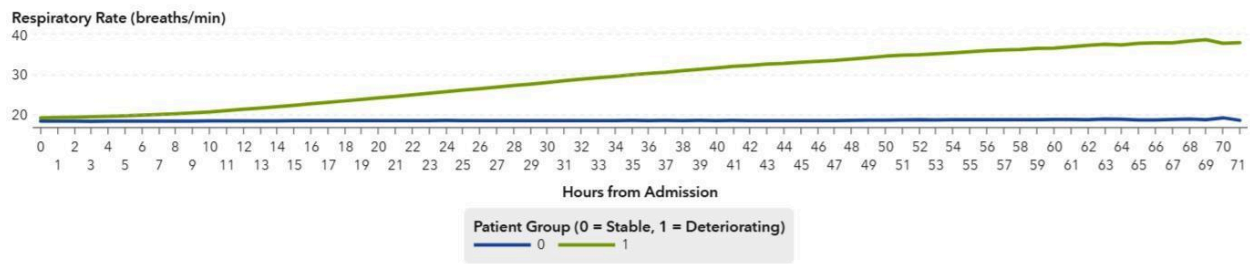


Figure 16. Respiratory Rate Trajectory by Deterioration Status — stable patients remain near 20 breaths/min; deteriorating patients rise to ~38 breaths/min by hour 71.

SpO2 Trajectory by Deterioration Status

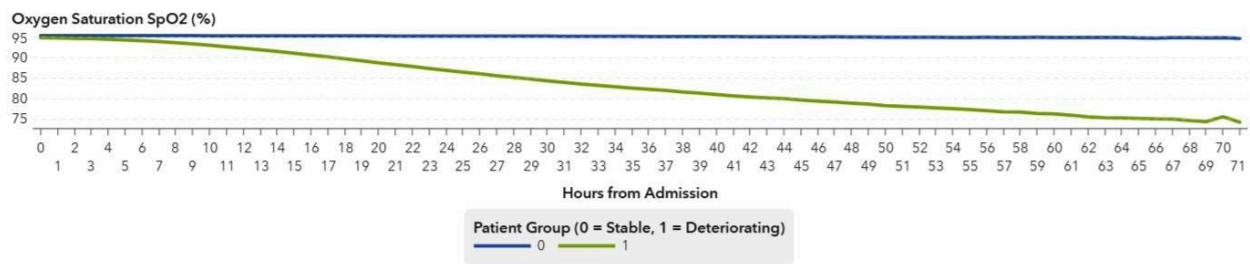


Figure 17. SpO2 Trajectory by Deterioration Status — stable patients remain near 95%; deteriorating patients decline steadily to ~75% by hour 71.

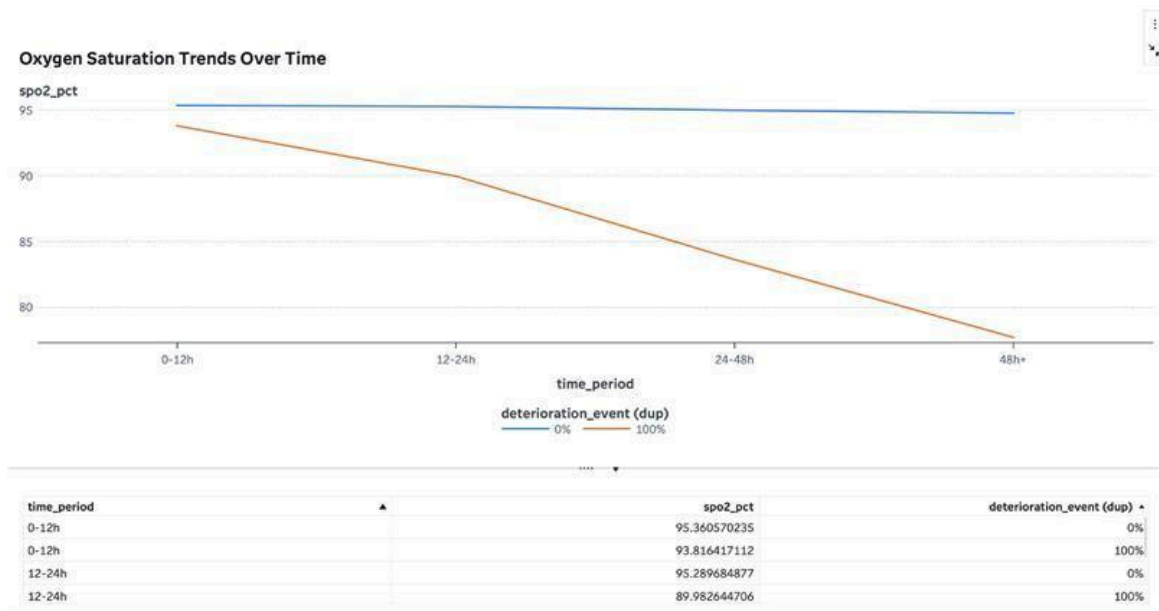


Figure 25. Oxygen Saturation Trends Over Time by deterioration status — stable patients (blue) hold steady at ~95% SpO2 across all time periods; deteriorating patients (orange) start at ~93.8% in the 0–12h window and drop progressively to ~78% by 48h+, demonstrating accelerating oxygenation decline.

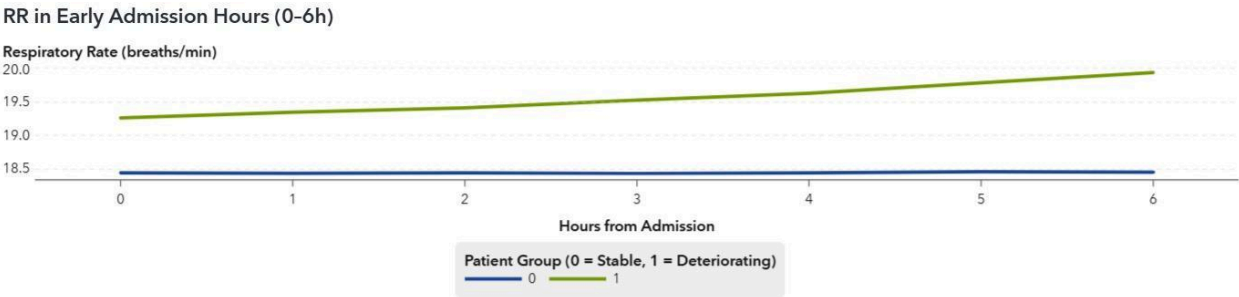


Figure 18. Respiratory Rate in Early Admission Hours (0–6h) — even in the first 6 hours, deteriorating patients show a slightly elevated baseline RR (~19.3 vs 18.5), providing an early warning opportunity.

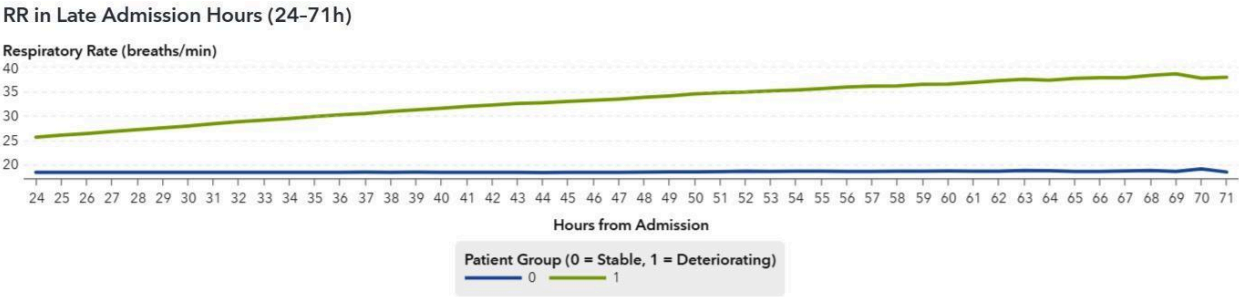


Figure 19. Respiratory Rate in Late Admission Hours (24–71h) — the divergence between stable and deteriorating patients becomes substantial in later hours.

5.6 Laboratory and Clinical Markers

Laboratory markers added important context beyond vital signs. Elevated CRP and infection-related patterns were associated with deteriorating patients. Creatinine spikes were more common among deteriorating patients, indicating kidney stress and broader organ instability. Abnormal hemoglobin also contributed to a more holistic understanding of deterioration risk.

Marker / Clinical Factor	Finding	Interpretation
Sepsis risk score	High-risk patients showed higher sepsis risk scores.	Supports infection-risk monitoring as part of deterioration assessment.
CRP / infection markers	Elevated CRP ranges appeared among deteriorating patients.	Inflammation may be associated with deterioration.
Creatinine	Elevated values were more common in deteriorating patients.	Signals kidney stress and systemic instability.
Hemoglobin	Abnormal hemoglobin patterns related to deterioration risk.	Adds value when combined with other signals.
Transfer status	Transferred patients showed higher clinical vulnerability.	Transferred patients may need closer monitoring due to care complexity.

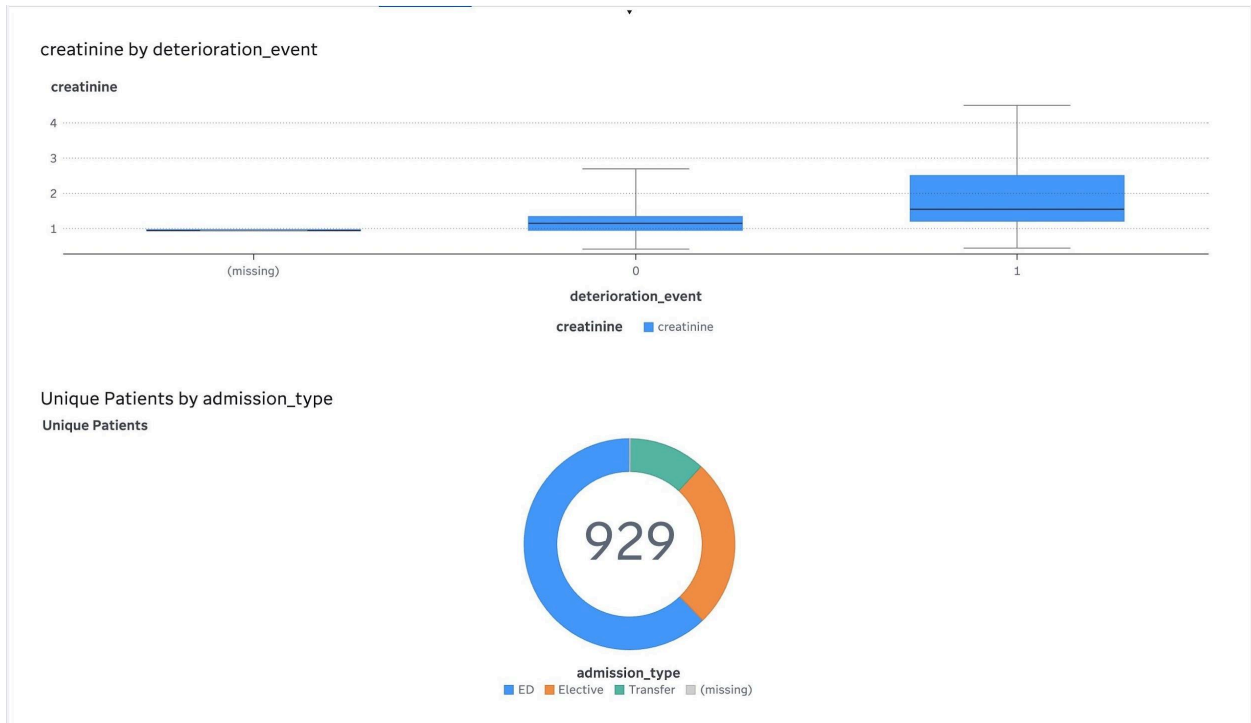


Figure 20. Creatinine by deterioration event (box plot) and unique patients by admission type (donut chart). Deteriorating patients (event = 1) show higher creatinine levels; ED admissions form the largest group (n = 929 total shown).

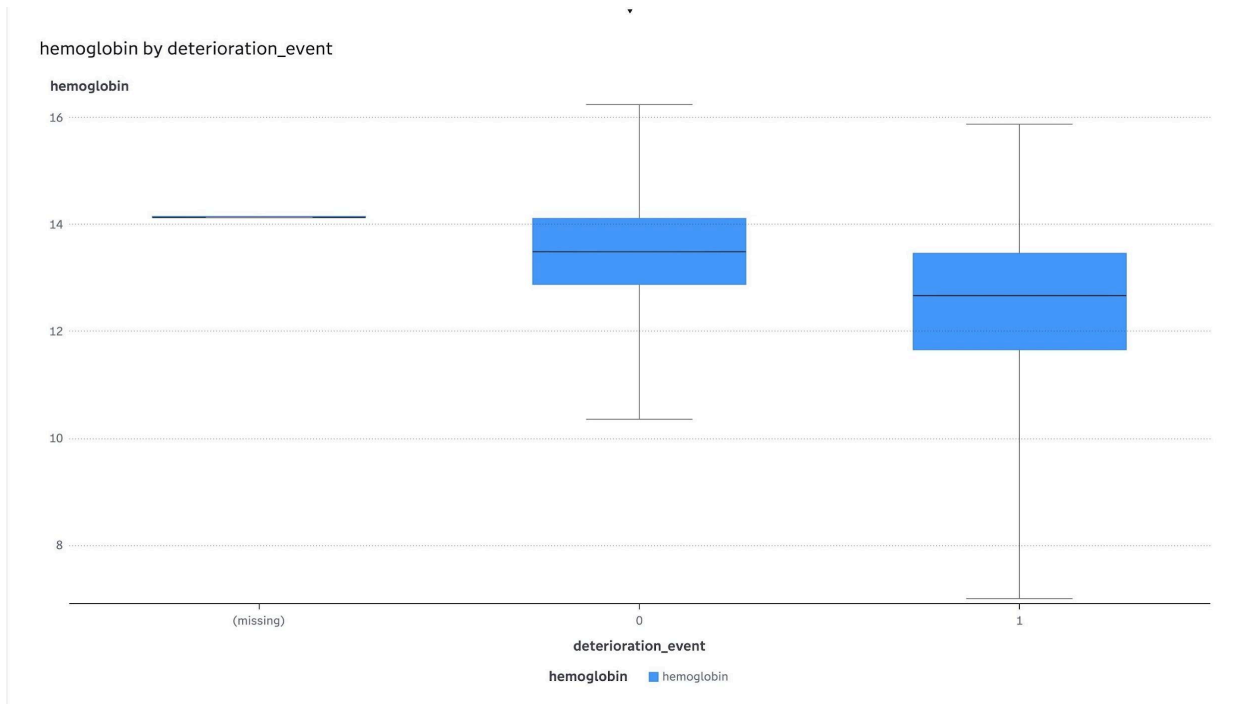


Figure 21. Hemoglobin by deterioration event — stable patients (event = 0) show a higher median hemoglobin (~13.5 g/dL) compared with deteriorating patients (event = 1, ~12.5 g/dL), consistent with anaemia-related vulnerability.

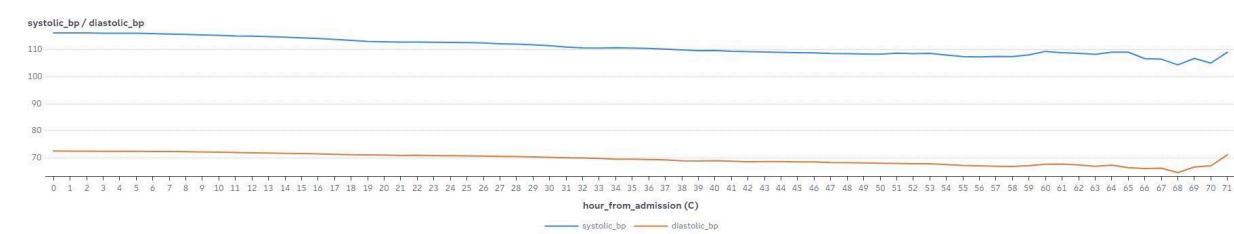


Figure 22. Temperature (°C) vs WBC Count by Deterioration Event — deteriorating patients (orange, event = 1) dominate the high WBC range (>15), and cluster toward higher temperatures, indicating inflammation and infection burden.

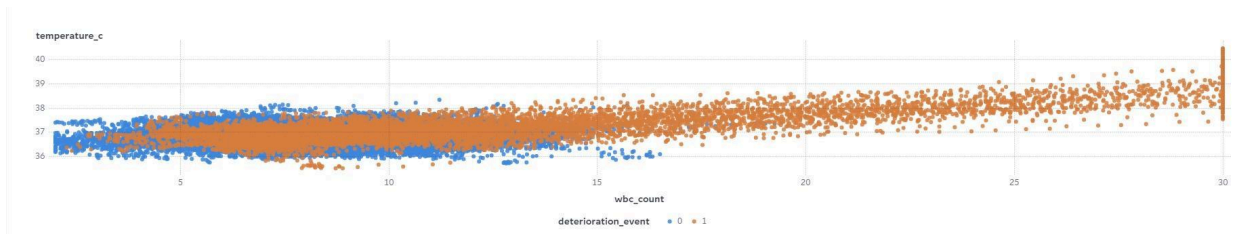


Figure 23. CRP Level Heatmap by Deterioration Event — stable patients (event = 0) concentrate at lower CRP values (<45); deteriorating patients (event = 1) extend across the full CRP range, including very high levels (>95), reflecting greater systemic inflammation.

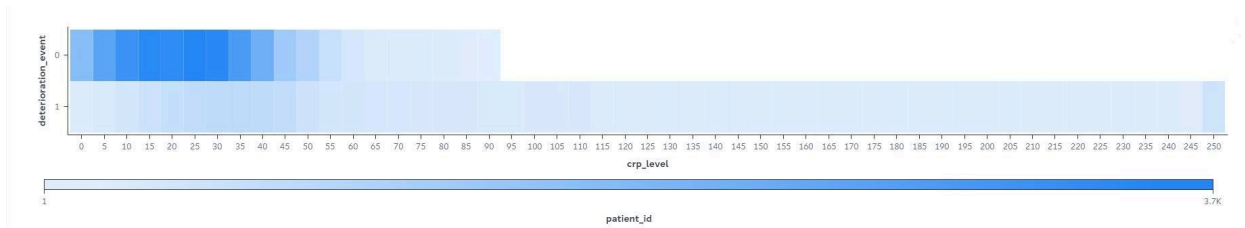


Figure 24. Lactate Heatmap by Hour from Admission — stable patients (event = 0) show consistently higher lactate intensity in early hours; deteriorating patients (event = 1) show a lighter, more dispersed pattern across the 72-hour window, suggesting metabolic stress accumulation over time.



Figure 27. Sepsis Risk Score Across Baseline Risk Groups — deteriorating patients (100%, orange) consistently show higher sepsis risk scores than stable patients (0%, pale) across all three baseline risk groups. High Risk deteriorating patients reach a sepsis score of 0.71 versus 0.49 for stable High Risk patients.

5.7 Dashboard and Storytelling Value

The SAS Visual Analytics dashboard approach is valuable because it connects patient safety questions with visual evidence. The analysis used KPI summaries, heat maps, bar charts, line charts, box plots, scatter plots, and treemaps to show both risk and workload. This helps stakeholders understand not only which factors matter, but also how to act on them.

The dashboard can support three practical decisions: which patients need closer monitoring, when monitoring should be intensified, and which clinical signals should trigger faster review.

6. Conclusion and Recommendations

The project demonstrates that predictive analytics can convert hospital monitoring data into practical early-warning insights. Deterioration was common enough to be clinically important, but not evenly distributed across all patients. Risk segmentation, admission profile, vital-sign trends, and laboratory markers together provide a stronger warning system than any single measure.

The strongest conclusion is that hospitals should use a combined monitoring workflow: baseline patient risk at admission plus live clinical signals during the first 24 to 72 hours. Respiratory rate, SpO2, heart rate, sepsis risk, creatinine, CRP, hemoglobin, oxygen support, mobility score, transfer status, and nurse alerts should be interpreted together.

6.1 Recommendations for Decision-Makers

- 1. Use risk segmentation at admission.** Classify patients into Low, Medium, and High Priority groups using age, comorbidity, admission type, and baseline risk score.
- 2. Prioritise High Priority patients during the first 24 hours.** These patients deteriorate earlier and should receive faster clinical review.
- 3. Combine patient profile with live clinical signals.** Baseline risk should be used together with respiratory rate, SpO2, heart rate, lactate, creatinine, oxygen support, nurse alerts, and mobility score.
- 4. Build an early-warning dashboard.** Display each patient's risk segment, warning signs, deterioration risk, and time since admission.
- 5. Support staff allocation.** Use dashboard insights to focus monitoring resources on patients with the highest risk and urgency.
- 6. Avoid single-indicator decision-making.** Heart rate, lab values, or risk scores alone are useful but insufficient; deterioration is best detected through combined trends.

Appendix A: Key Metrics

Metric	Value / Finding
Total patients	10,000
Hourly observations in final analysis	418,000
Patients deteriorating	19%
Near-term risk	3.1%
High Priority patients	12.5% of patients
High Priority deterioration rate	Approximately 29.4%
Low Risk deterioration rate	Approximately 12.6%
High Risk deterioration rate	Approximately 26.6%
Highest-risk segment	80+ patients with very high comorbidity (~37% deterioration risk)
Median deterioration time (High Priority)	21.5 hours
Median deterioration time (Low Priority)	26 hours
Average heart rate – Stable patients	~84 BPM
Average heart rate – Deteriorating patients	~108 BPM
Respiratory rate trend (Deteriorating)	~20 → 38 breaths/min over 72 hours
SpO2 trend (Deteriorating)	~94% → 74% over 72 hours